Fiscal Year:	FY 2025	**Task Last Updated:**	FY 05/04/2025
PI Name:	Mason, Christopher Ph.D.		
Project Title:	The Impact of Spaceflight and Radiation on Clonal Hematopoiesis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10021-5663	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	05/01/2023	**End Date:**	04/30/2026
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Eliah Overbey has now left to start her own laboratory at UATX.		
COI Name (Institution):	Bailey, Susan Ph.D. (Colorado State University) De Vlaminck, Iwijn Ph.D. (Cornell University) Elemento, Olivier Ph.D. (Weill Medical College of Cornell University) Hassane, Duane Ph.D. (Weill Medical College of Cornell University) Levine, Ross MD (Sloan Kettering Institute for Cancer Research)		
Grant/Contract No.:	80NSSC23K0832		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Clonal hematopoiesis (CH) occurs when blood cells harboring an advantageous mutation propagate faster than others. These mutations can confer a risk for certain hematological cancers and cardiovascular disease. However, little is known about the methods for reducing these clones, nothing is known about CH in astronauts, and there is little information about the proper way to understand and stratify the risk of these mutations that are found in normal individuals. The goal of this proposal is to delineate the risk of CH for astronauts and other patients using improved, novel methods that map nucleic acids isolated from peripheral blood and urine. We hypothesize that radiation imparts a selective advantage, specifically in hematopoietic cells with DNA Damage Response mutations, and that spaceflight duration and radiation dose can drive and mediate this selection process. We will use a cohort of >1,100 patients from Weill Cornell Medicine (WCM) with both blood and urine, where 11% already show CH, as well as existing data from healthy cohorts (n=20,000) from the Englander Institute for Precision Medicine (EIPM) and Memorial Sloan Kettering Cancer Center (MSKCC) (n=25,000). We will use established methods for purifying and extracting DNA, using protocols developed for the NASA Twins Study and our published EIPM PreCISE-1 test for CH. This includes 100+ genes involved in clonal hematopoiesis, cancer, and cardiovascular diseases (CVD), sequenced to a depth of >2000x coverage (with unique molecular barcodes), giving >95% sensitivity for variant allele frequencies (VAF) > 1% (about 1/50 cells). The presence of CH can inform long-term CVD and cancer risk, and thus can serve as a useful metric for astronauts preparing for long-term missions. Also, CH confers a mutation-specific inflammatory risk, which can potentiate post-flight or post-radiation morbidity and be compared to other risk factors. Thus, identifying and managing CH represents a crucial unmet need for astronaut health, radiation risk stratification, and guiding oncology patients more broadly.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We completed analysis of the WCM CH cohort and also the Inspiration4 crew, which was published in June of 2024 the Space Omics and Medical Atlas (SOMA) package. We are now processing data from other crews such as Polaris Dawn and other missions.
Task Progress:	We have completed the targeted CH assay for blood samples in the Inspiration4 cohort, and completed a follow-up analysis of the whole blood samples from other cohorts. This has led to two publications specifically focused on the questions of CH and a package of papers published in June 2024 in Nature, Nature Communications, and related Nature partner journals. This work was also featured on the cover of Nature and hosts a dedicated landing page for the Space Omics and Medical Atlas (SOMA): https:// Specifically, in the I4 cohort (Garcia-Medina et al, 2024), we found no relationship between spaceflight and increased CHIP-related genetic abnormalities in the scale of three months after spaceflight. Telomere elongation occurred during spaceflight and shortened after return to Earth. Cell-free DNA analysis revealed increased immune cell signatures post-flight. While TET2 exhibited a missense mutation chr4:g.105275662G>T at a VAF ~ 0.3 in C003, no significant clonal hematopoiesis of indeterminate potential (CHIP) or whole-genome instability was observed. The long-term gene expression changes across immune cells suggested cellular adaptations to the space environment persisting months post-flight. However, putative mutations in CH-linked genes were not found at disproportionate allele frequencies post-flight, nor were de novo mutations seen in the post-flight samples. Although our findings suggest that no pathological genomic alterations occurred as a function of spaceflight in the I4 cohort, further longitudinal tracking of the astronauts will help us discern the long-term effects, if any, of short-duration missions. Fortunately, one of the same crew members from I4 also flew on Polaris Dawn (PD), and we are now processing these samples with improved methods to delineate a long-term differential for CH changes. These data provide a unique opportunity to measure longitudinal changes in CH, which can be leveraged for a better understanding of long-term risk and clonal evolution. In terms of mentorship, publications supported by this grant led to one of our lab's scientists (Dr. Eliah Overbey) getting a faculty position at the University of Austin in the fall of 2024 and another member of the lab (Dr. Jiwoon Park) getting a post-doctoral fellowship in the George Church lab at Harvard. This will extend the space medicine work to a new lab, which will continue to collaborate with our team and use these data, as part of the SOMA groups. Also, these funds helped several projects and informed best practices in the large Inspiration4/JAXA/NASA package in Nature, which is featured among a wide range of 22 publications across 2023-2025.
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